

Caellum Yip Hoi-Lee

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EDUCATION

University of Toronto

Honours Bachelor of Science — Computer Science Program

- Planning to enroll in the Computer Science Specialist Program.

Toronto, ON

Sep 2025 – Apr 2029

EXPERIENCE

Software Engineer, Chatforce

Chatforce Office

February 1st, 2026 – Present

Toronto, ON

- Completely overhauled AI game development pipeline, singlehandedly implementing a vision-driven multi-agent architecture inspired by **OpenGame** research and employing a custom affordance model.
- Developed the Chatforce Creator Economy and built a Creator Analytics Dashboard using **Angular** and **TailwindCSS** that pulls from S3 buckets
- Used (**Angular**, **FastAPI**) to create key platform features, including game versioning, taglines, and patch notes
- Developed a **FastAPI** rate-limiting system with a Redis deduplication cascade, utilizing **Pydantic** validation and IPv6 prefix truncation to prevent play-count manipulation.

Software Developer, U of T Blueprint

University of Toronto

Sep 2025 – May 2026

Toronto, ON

- Collaborating with a non-profit organization developing software for social change.
- Developed JWT Authentication/Middleware, Admin Dashboard, and CSV Data Management system with **Django REST** and **React** for the Museum of Art and Digital Entertainment in California.

Front End Developer, U of T Boxing Club

University of Toronto

Jan 2026–March 2026

Toronto, ON

- Developed a website for the U of T Boxing Club with **React**, **Next.js**, and **Tailwind CSS**.
- Implemented a **CMS** with Google Sheets and Google Calendar APIs to dynamically manage merchandise, classes, and event schedules for **40+ active members**.

Owner and Software Lead, Bronze Bat Studio

Independent Game Studio

Aug 2023 – Present

- Founded and led a small game development studio focused on **Lua** and **C#**.
- Published titles achieving over **3 million plays** and generating approximately **\$2,000** in revenue on Steam and Roblox.

Head of Programming / Co-founder, RHHS Game Design

Richmond Hill High School

Sep 2023 – June 2025

Richmond Hill, ON

- Developed and taught interactive Unity lessons in **C#** guiding members through creating a Flappy Bird-style game.
- Implemented gameplay mechanics in lessons to teach programming concepts such as conditionals, functions, and Unity physics.

SKILLS AND CERTIFICATIONS

Programming Languages: Java, Python, C#, C++, SQL, JavaScript, Lua, R; strong in DSA and OOP

Web & Mobile Development: React, React Native, Next.js, Django REST Framework, FastAPI, Streamlit, HTML/CSS

AI & Machine Learning: PyTorch, LangChain, ChromaDB, Scikit-learn, Hugging Face, Diffusion Models (SDXL), LoRA Fine-tuning, RAG, MiroShark/MiroFish, Gemini 2.5 Pro, Veo 2, OpenRouter, Meshy, Moorcheh AI, Kaggle

Tools & Platforms: Docker, Git, Linux, AWS Trainium, Postman, Playwright, Cloudinary, Composio, CyStack, Polarity, Firebase, Jupyter Notebook, Quarto MD

Database & Concepts: PostgreSQL, Supabase, Redis, Neo4j, SQLite3, Vector Search (Redisearch/HNSW), WebSockets, JSON Web Tokens (JWT), Server-Sent Events (SSE)

Languages: English, French, Spanish

Certifications: Ontario Secondary School Diploma (AP Student, 97% Avg), Royal Conservatory of Music — Piano Level 8

Awards: RHHS English Subject Course Award; RHHS Gold Band Certificate of Merit

- Paper Cuts (Cal Hacks AI Hackathon)** | *Python, PyTorch, FastAPI, Redis, AWS Trainium, JavaScript* Jun 2026
- Won **1st Place Overall (\$5,000)**, **Best UI/UX**, and **Top 10 Finalist** for a real-time, draw-to-play game that turns children's doodles into playable game assets in seconds.
 - Trained a custom **image-to-image diffusion** model (**SDXL + InstructPix2Pix**) on **AWS Trainium** to convert rough sketches into clean, game-ready sprites.
 - Built a cascading recognition pipeline pairing a custom **CNN**, an open-vocabulary **VLM** fallback, and **retrieval-augmented recognition** over a **Redis (RediSearch HNSW)** vector memory that sharpens as more users draw.
 - Built **Moose**, a **neuro-symbolic** mechanic composer combining a **LoRA**-tuned model with a safe-by-construction rule-graph to map any drawing to a valid, balanced mechanic with elemental interactions.
 - Built the game engine from scratch on **HTML5 Canvas** with **WebSocket** phone-as-controller multiplayer and AI-generated cinematic finishers (**Pika**).
- Auctopus (SOON Hackathon)** | *Next.js, Gemini 2.5 Pro, Veo 2, Neo4j, Cloudinary* May 2026
- Won **First Place** in the **Cloudinary** and **Polarity** sponsor tracks for developing an end-to-end automated product validation and prototyping pipeline.
 - Engineered an asynchronous 8-stage execution chain within a Next.js Route Handler, utilizing **SSE** to stream live generation progress to the UI.
 - Orchestrated a multi-model workflow integrating **Gemini 2.5 Pro** for data-driven brand research, **Veo 2** for video generation, and a **Neo4j-backed** MiroShark architecture for multi-persona audience simulation.
 - Leveraged **Cloudinary's** **CDN** and the **Meshy API** to dynamically generate, transform, and deliver interactive 3D GLB/USDZ models for Apple AR Quick Look and 360-degree product stills.
 - Automated enterprise data distribution via **Composio** to push analytics to Slack and Notion, while utilizing **CyStack** for secret injection and **Polarity** for AI-driven PR reviews and automated test generation.
- Minutes (GenAIGenesis Hackathon)** | *Python, FastAPI, LangChain, ChromaDB, Docker, Supabase* Mar 2026
- Awarded **Runner-Up for Google Best AI for Community Impact** for developing an autonomous civic agent that broadcasts municipal legislative updates to social media.
 - Engineered a data ingestion pipeline utilizing **Playwright** and **BeautifulSoup4** to bypass complex authentication systems and scrape undocumented REST APIs from the Toronto municipal portal.
 - Implemented a RAG architecture using **LangChain** and **ChromaDB** to process scraped PDFs and generate structured, jargon-free policy summaries through OpenRouter.
 - Built an automated DM workflow using the **Instagram Graph API** and RLS-encrypted **PostgreSQL** to securely process user petitions and draft emails to local representatives.
- Apogee AI** | *Python, Streamlit, Scikit-learn, Gemini API, Moorcheh AI* DeltaHacks 12, Jan 2026
- Used **Louvain's clustering/modularity algorithm** and **Pyvis** to generate a 3D node map of vectorized news articles clustered by subtopics.
 - Scraped articles with **Newspaper3k + NewsAPI**, vectorized text with lightweight Hugging Face transformers, and served/deployed results with Streamlit + HTML/CSS.
 - Implemented **Moorcheh AI retrieval-augmented queries** and **Gemini API** to deliver a chatbot with memory and context-aware responses sourced from scraped articles.
 - Deployed further transformers to filter content by political spectrum and emotion, utilizing **named-entity recognition** to plot articles on an interactive global map.
- Twin Universe Space Simulator** | *Lua* Bronze Bat Studio, Aug 2023 – Present
- Developed a space exploration game with rocket launches, progression systems, and multiplayer missions.
 - Created interactive environments and engaging mechanics leading to over **2.8 million player visits**.
- Trace Stock Market Predictor** | *Python* Dec 2025
- Used Tensor Flow to develop and train a **Long Short-Term Memory (LSTM)** network to predict stock price trends from historical market data.
 - Developed a data ingestion pipeline using the **Alpha Vantage API** to clean 20+ years of global OHLC data into Pandas DataFrames.
- Genetic Code/Cellular Automata Experiments** | *R* Sep 2025 – Dec 2025
- Performed computational experiments to assess SGC mutation tolerance and the impact of augmentors in **Wolfram Rule 110 cellular automata**.
 - Compiled Quarto/Jupyter notebooks documenting findings.